

Weston Fleming

Postdoctoral Scientist

Howard Hughes Medical Institute, University of Washington

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Education

- 2015 - 2022** **Princeton University**
Ph.D. in Neuroscience
Thesis: "Measurement and Manipulation of Subcortical Neuromodulatory Populations during Reward Learning"
- 2010 - 2014** **Duke University**
B.Sc. in Neuroscience, Graduation with Distinction

Research positions

- 2022 - present** **University of Washington, HHMI, Postdoctoral Scientist**
Advisors: Richard Palmiter, Ph.D., Garret Stuber, Ph.D.
- 2015 - 2022** **Princeton University, Graduate Student**
Advisor: Ilana Witten, Ph.D.
- 2013 - 2014** **Duke University, Undergraduate Research Assistant**
Advisor: Cynthia Kuhn, Ph.D.

Publications

Preprints

- 2026** **Fleming, W.**, Pauli, J.L., Ishii, K.K., Trzeciak, M., Burke, C.T., Park, S., Gordon, A.G., Zweifel, L.S., Palmiter, R.D., Stuber, G.D. A cholinergic eligibility trace facilitates amygdala plasticity in flavour avoidance learning. *bioRxiv*. 2026. [Link](#).

Published articles

- 2025** Simon, R.C., **Fleming, W.T.**, Briones, B.A., Trzeciak, M., Senthikumar, P., Ishii, K.K., Hjort, M.M., Martin, M.M., Hashikawa, K., Sanders, A.D., Golden, S.A., Stuber, G.S. Opioid-driven disruption of the septum reveals a role for neurotensin-expressing neurons in withdrawal. *Neuron*. 2025. [Link](#).
- 2024** Minakuhi, T., Guthman, E.M., Acharya, P., Hinson, J., **Fleming, W.**, Witten, I.B., Oline, S.O., Falkner, A.L. Independent inhibitory control mechanisms for aggressive motivation and action. *Nature Neuroscience*. 2024. [Link](#).
- 2023** Cox, J., Minerva, A., **Fleming, W.**, Zimmerman, C. A., Hayes, C., Bandi, A., Ornelas, S., Parker, N.F., Witten, I.B. A neural substrate of sex-dependent modulation of motivation. *Nature Neuroscience*. 2023. [Link](#).

- 2022** **Fleming, W.**, Lee, J., Briones, B., Bolkan, S., Witten, I.B. Cholinergic interneurons modulate cocaine extinction through shared plasticity mechanisms across medium spiny neuron subtypes. *Cell Reports*. 2022. [Link](#).
- Briones, B.A., Pitcher, M.*, **Fleming, W.***, Libby, A., Diethorn, E., Haye, A.E., MacDowell, C. J., Zych, A.D., Waters, R.C., Buschman, T.J., Witten, I.B., Gould, E. Increased perineuronal nets surrounding parvalbumin interneurons in the dorsomedial striatum are associated with excessive repetitive behaviors. *Biological Psychiatry Global Open Science*. 2022. *Equal contributions. [Link](#).
- 2021** **Fleming, W.***, Jewell, S.*, Enghelard, B., Witten, D.M., Witten, I.B. Inferring spikes from calcium imaging in dopamine neurons. *PLOS One*. 2021. *Co-first authors. [Link](#).
- 2020** Choi, J.Y., Jang, H.J., Ornelas, S., **Fleming, W.**, Fürth, D., Au, J., Bandi, A., Engel, E.A., Witten, I.B. A comparison of dopaminergic and cholinergic populations reveals unique contributions of VTA dopamine neurons to short-term memory. *Cell Reports*. 2020. [Link](#).
- Cai, L.X., Pizano, K., Gundersen, G.W., Hayes, C.L., **Fleming, W.**, Holt, S., Cox, J.M., Witten, I.B. Distinct signals in medial and lateral VTA dopamine neurons modulate fear extinction at different times. *eLife*. 2020. [Link](#).
- 2019** Engelhard, B., Finkelstein, J., Cox, J., **Fleming, W.**, Koay, S.A., Thiberge, S., Tank, D.W., Witten, I.B. Specialized and spatially organized coding of sensory, motor, and cognitive variables in midbrain dopamine neurons. *Nature*. 2019. [Link](#).
- Fleming W.**, Jones Q., Chandra U., Saini A., Walker Q.D., Francis R., Ocampo G., Kuhn, C. Withdrawal from brief repeated alcohol treatment in adolescent and adult male and female rats. *Alcoholism Clinical & Experimental Research*. 2019. [Link](#).

Honors and Awards

- 2026-** NIH Pathway to Independence Award (K99/R00, pending; Impact Score = 19)
Title: "A role for central amygdala acetylcholine in conditioned taste aversion"
Advisors: Richard Palmiter & Garret Stuber
- 2025** Selected speaker, Amygdala GRC
- 2017-2022** NSF Graduate Research Fellowship
Advisor: Ilana Witten
- 2017** Sigma Xi Research Honor Society nomination
- 2013** Duke Neuroscience Program of Research Fellow
- 2010-2014** Kids' Chance of South Carolina Scholar
- 2010** Byrd Scholar

2010 Carolina Scholar, University of South Carolina

Teaching and mentorship

2024- Lecturer, “2-photon Pre-processing and Analysis.” UW NAPE Center Imaging & Genetics Workshop. University of Washington.

2024-2026 Mentor, Undergraduate research assistant (Ruohe Liu). University of Washington.

2023 Mentor, Rotation graduate student (Monica Tschang). University of Washington.

2019 Mentor, PNI Summer Internship Program (A Yang, University of Wisconsin – Green Bay). Princeton University.

2017-2018 Mentor, ReMatch (Lucas Lee). Princeton University.

2017 Teaching Assistant, Introduction to Neuroscience. NEURO 175. Princeton University.

2016 Teaching Assistant, Introduction to Neuroscience. NEURO 201. Princeton University.

Invited talks

2026 **Fleming, W.** UW NAPE Center Seminar Series. Seattle, WA. USA. Planned.

Fleming, W. CoNECTOME 2026. Seattle, WA. USA.

2025 **Fleming, W.** Gordon Research Seminar on Amygdala Function in Emotion, Cognition, and Disease. Barcelona, Spain.

Fleming, W. Conference on Stress, Pain, Emotion, Addiction, and Reward. Friday Harbor, WA. USA.

2022 **Fleming, W.** UW NAPE Center Computing Group. University of Washington. Seattle, WA. USA.

2021 **Fleming, W.** Princeton Neuroscience Institute In-house Seminar. Princeton, NJ. USA.

Fleming, W. Johns Hopkins University. Baltimore, MD. USA. Virtual.

Fleming, W. University of California – San Diego. San Diego, CA. Virtual.

Fleming, W. University of Washington. Seattle, WA. USA. Virtual.

2020 **Fleming, W.** Princeton Neuroscience Graduate Research Seminar. Princeton, NJ. USA.

2019 **Fleming, W.** Gordon Research Seminar on Modulation of Neural Circuits and Behavior. Les Diablerets, Switzerland.

Conference Abstracts / Posters

2026 **Fleming, W.**, Pauli, J., Ishii, K., Trzeciak, M., Palmiter, R., Stuber, G. CoNECTOME 2026. Seattle, WA. USA.

Fleming, W., Pauli, J., Ishii, K., Trzeciak, M., Palmiter, R., Stuber, G. UW NAPE Symposium. Seattle, WA. USA.

Fleming, W., Pauli, J., Ishii, K., Trzeciak, M., Palmiter, R., Stuber, G. HHMI Science Meeting. Ashburn, VA. USA.

2025 **Fleming, W.**, Pauli, J., Ishii, K., Trzeciak, M., Palmiter, R., Stuber, G. Gordon Research Conference on Amygdala Function in Emotion, Cognition, and Disease. Barcelona, Spain.

2021 **Fleming, W.***, Jewell, S. *, Engelhard, B., Witten, D.M., Witten, I.B. Virtual Dopamine (ViDA) Conference. Virtual. *Co-first authors.

2020 **Fleming, W.***, Jewell, S. *, Engelhard, B., Witten, D.M., Witten, I.B. Virtual Dopamine (ViDA) Conference. Virtual. *Co-first authors.

2019 **Fleming W.**, Lee J., Witten I.B. Gordon Research Conference on Modulation of Neural Circuits and Behavior. Les Diablerets, Switzerland.

2018 **Fleming W.**, Lee J., Witten I.B. Society for Neuroscience. San Diego, CA. USA.